

Real Life Applications of Imaginary Numbers

Complex numbers enter into studies of physical phenomena in ways that most people can't imagine

- For example: A differential equation, with coefficients like the a , b , and c in the quadratic formula, that models how electrical circuits or forced spring/damper systems behave. The movement of the shock absorber of a car as it goes over a bump is an example of the latter. The behavior of the differential equations depends upon whether the roots of a certain quadratic are complex or real. If they are complex, then certain behaviors can be expected. These are often just the solutions that one wants.
- Closely related to the electrical engineering example is the use of complex numbers in signal processing, which has applications to telecommunications (cellular phone), radar (which assists the navigation of airplanes), and even biology (in the analysis of firing events from neurons in the brain).
- In modeling the flow of a fluid around various obstacles, like around a pipe, complex analysis is very valuable for transforming the problem into a much simpler problem. When everything from large structures of riveted beams to economic systems are analyzed for resilience, some very large matrices are used in the modeling.
- In everyday use, industrial and university computers spend some fraction of their time solving polynomial equations. The roots of such equations are of interest, whether they are real or complex.

Applications to potential flow

Potential Flow

Potential Flow

- Suppose that a fluid flow over the complex plane $\mathbb{C}$ and that the velocity at the point $z = x + iy$ be given by the velocity vector

$$\mathbf{V}(x, y) = u(x, y) + iv(x, y). \quad (1)$$

- We assume that the velocity does not depend on time and $u(x, y)$ and $v(x, y)$ have continuous first order partial derivatives.
- The divergence vector field of $\mathbf{V}$ is given by

$$\operatorname{div} \mathbf{V} = u_x(x, y) + v_y(x, y)$$

and is a measure to the extent to which the velocity diverges near the point.

Potential Flow

- We only consider fluid flows that have divergence zero. Such fluids are called **irrotational** fluid flows. This means that the flow of fluid through any simple closed contour is identically zero.
- The equation $u_x(x, y) + v_y(x, y) = 0$ gives the equation of continuity of the flow. If u_x and v_y satisfy the Cauchy Riemann equations then by the continuity of partial derivatives we have the corresponding function $f = u + iv$ as analytic function.

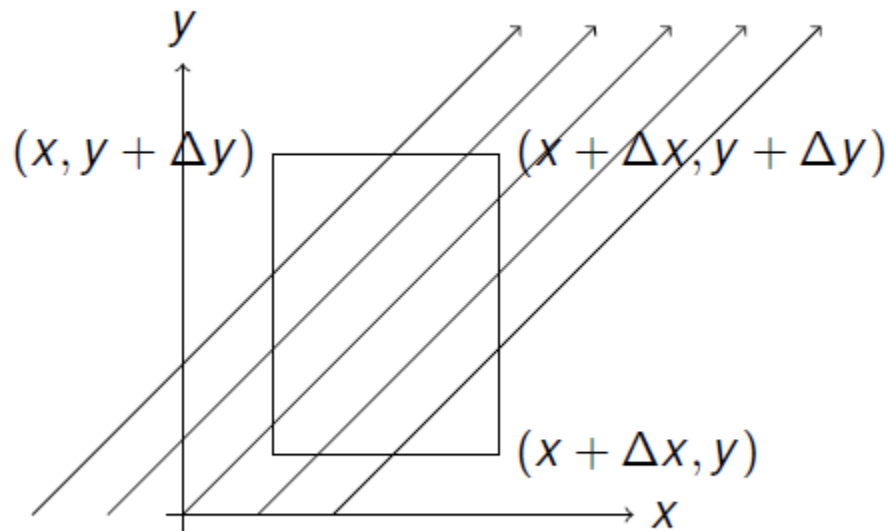


Figure : 2-D fluid flow

Potential Flow

- Now this analytic function has an antiderivative $F(z)$ such that

$$f(z) = F'(z), \quad \text{where} \quad F(z) = \phi(x, y) + i\psi(x, y).$$

- This $F(z)$ is called complex potential and has the property that

$$\overline{F'(z)} = \phi_x(x, y) - \psi_x(x, y) = u + iv = \mathbf{V}(x, y).$$

- Thus

$$\phi_x = u \quad \text{and} \quad \phi_y = -\psi_x = v.$$

- Also $\text{grad}\phi = u + iv = \mathbf{V}(x, y)$.

Potential Flow

- (i) $\phi(x, y)$ is called velocity potential and the curves $\phi(x, y) \equiv c_1$ (constant) are called Equipotentials of the fluid flow.
- (ii) $\psi(x, y)$ is called stream line and the curves $\psi(x, y) \equiv c_2$ (constant) are called Stream lines of the fluid flow.

Potential Flow

Example

Let the complex potential of the fluid be given by $F(z) = (\alpha + i\beta)z$. Then

- (i) Velocity vector is $\phi(x, y) = ax - by$ and $ax - by = c_1$, are equipotentials.
- (ii) Stream line is $\psi(x, y) = bx + ay$ and $bx + ay = c_2$, are the stream line flows. These lines are parallel and makes an angle $\theta = \tan^{-1}(b/a)$.

Example

For the complex potential $F(z) = \frac{B}{3}z^3$, $B > 0$. Velocity potential is

$$\phi(x, y) = \frac{Bx}{3}(x^2 - 3y^2)$$

and the stream line is

$$\psi(x, y) = \frac{By}{3}(3x^2 - y^2).$$